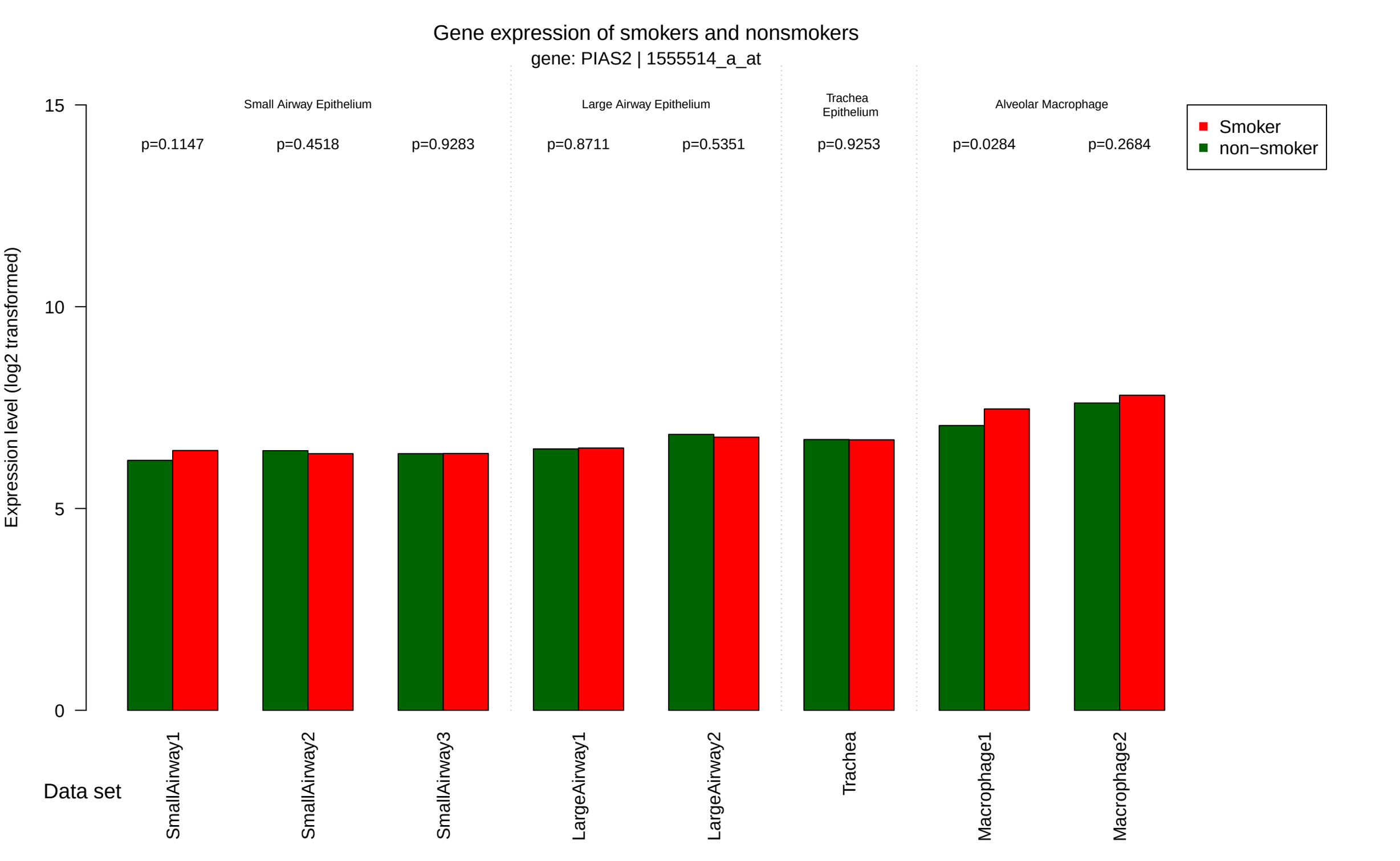
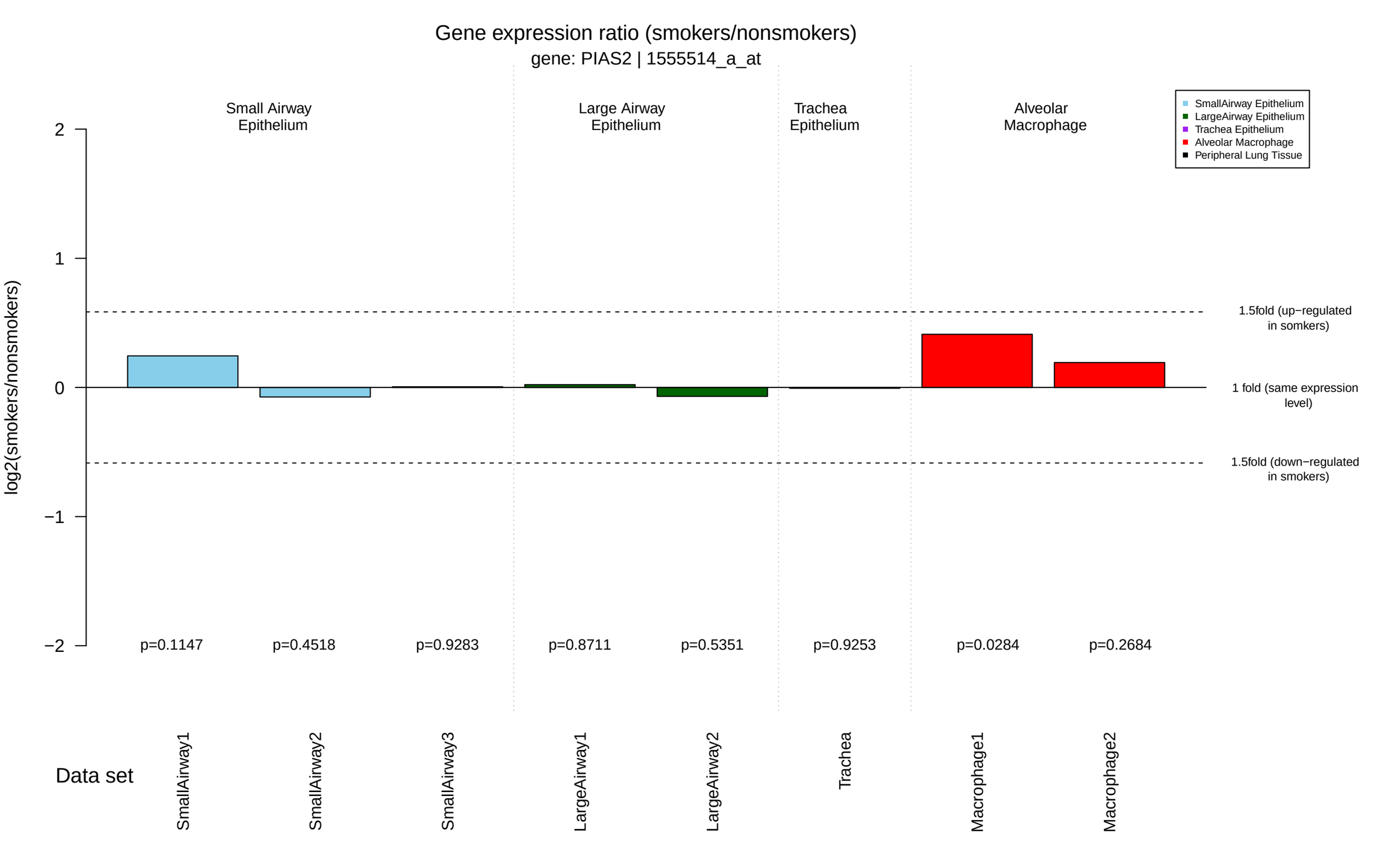
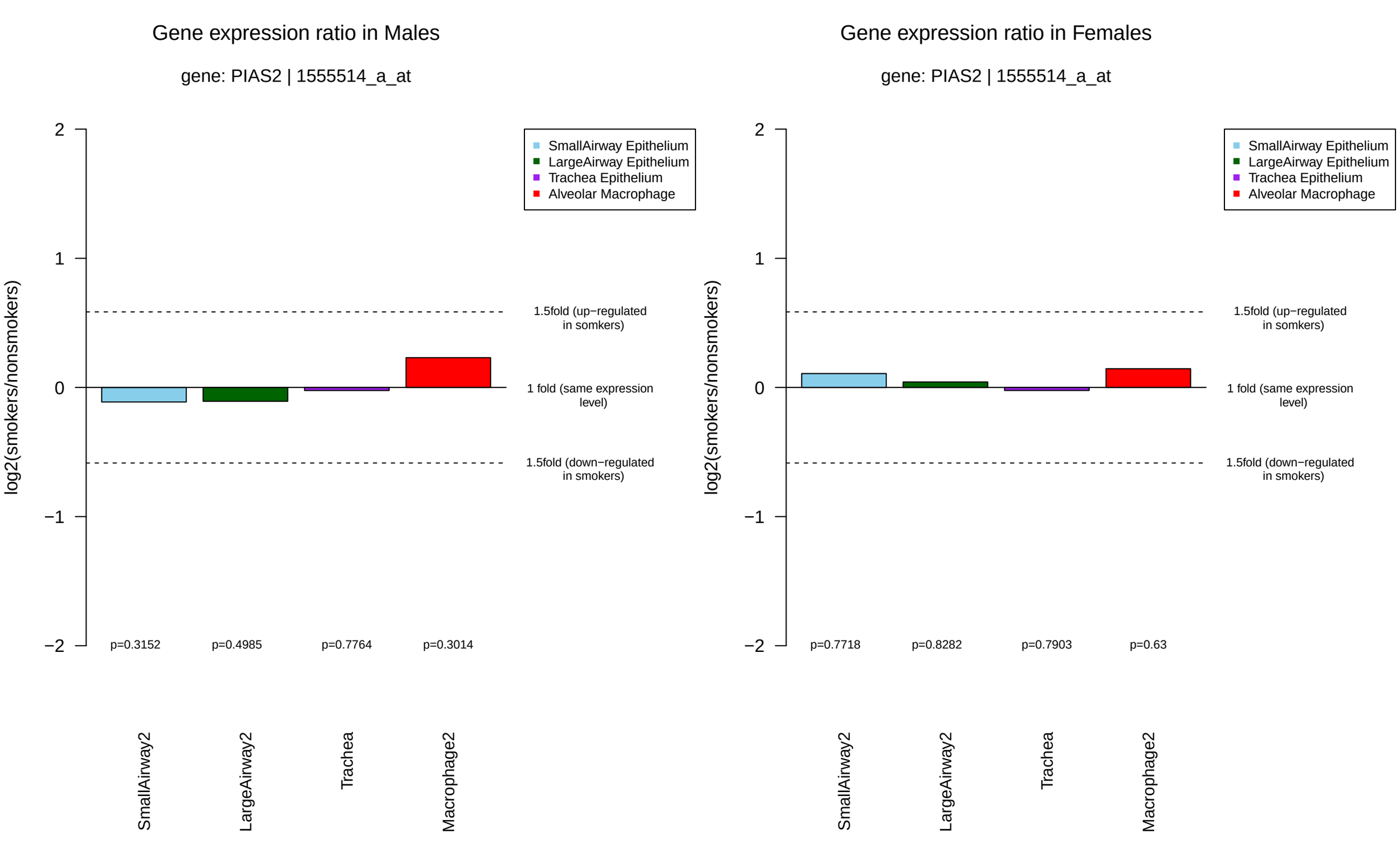
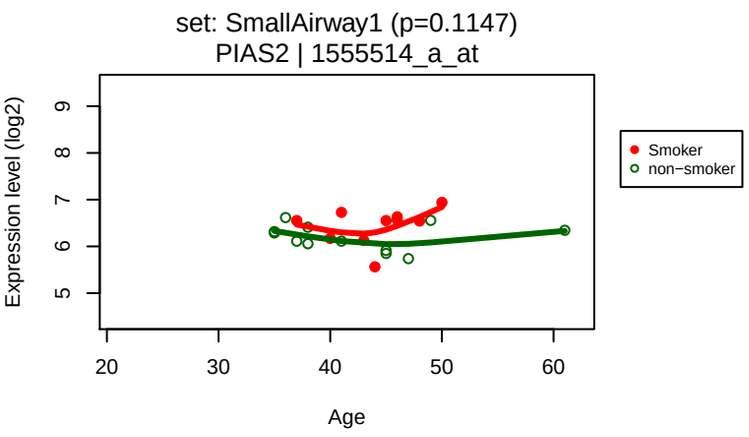
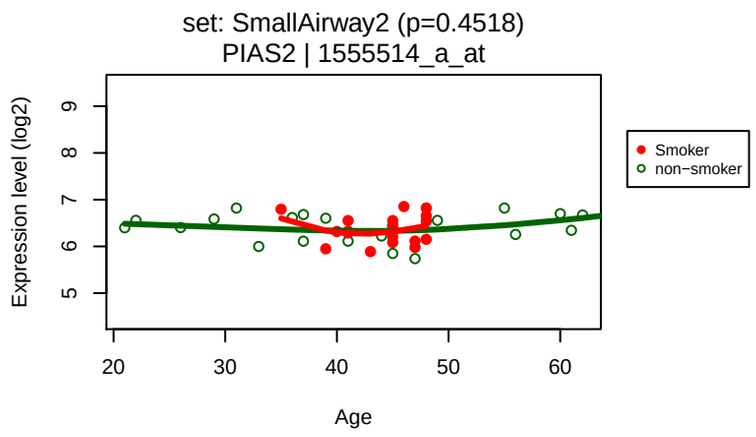
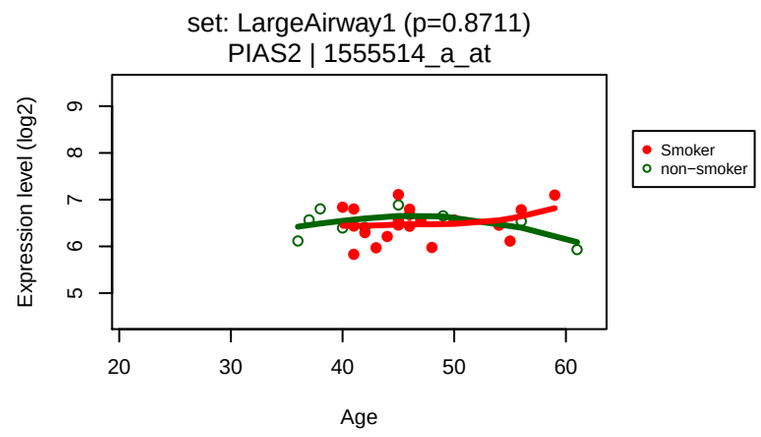

Gene expression of smokers and nonsmokers
gene: PIAS2 | 1555514_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1147
p=0.4518
p=0.9283
p=0.8711
p=0.5351
p=0.9253
p=0.0284
p=0.2684
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

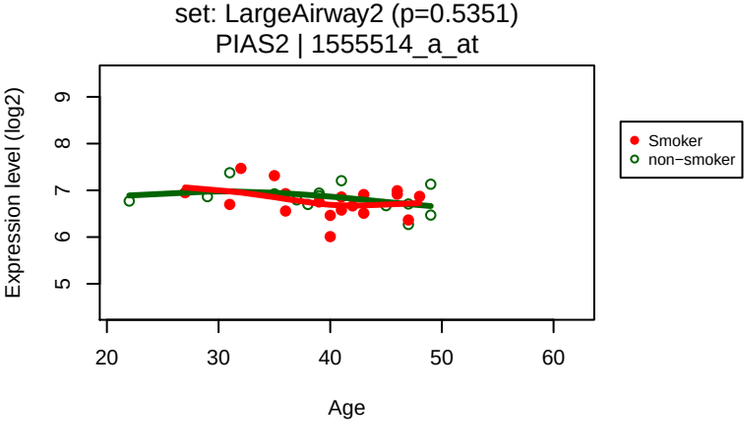

Gene expression ratio (smokers/nonsmokers)
gene: PIAS2 | 1555514_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1147
p=0.4518
p=0.9283
p=0.8711
p=0.5351
p=0.9253
p=0.0284
p=0.2684
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

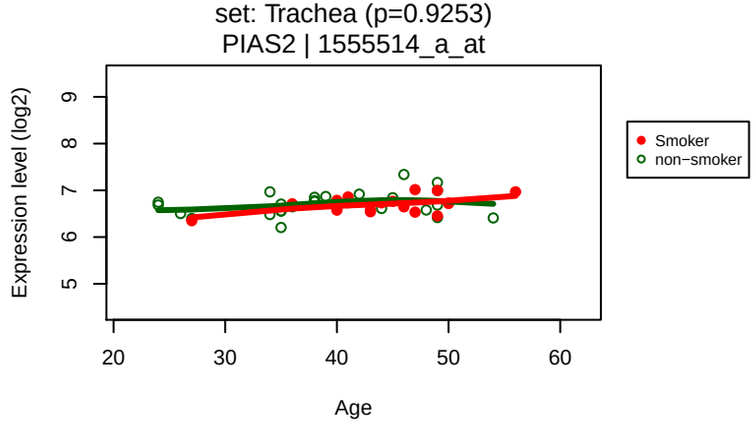

Gene expression ratio in Males
gene: PIAS2 | 1555514_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3152
p=0.4985
p=0.7764
p=0.3014
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PIAS2 | 1555514_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7718
p=0.8282
p=0.7903
p=0.63
SmallAirway2
LargeAirway2
Trachea
Macrophage2

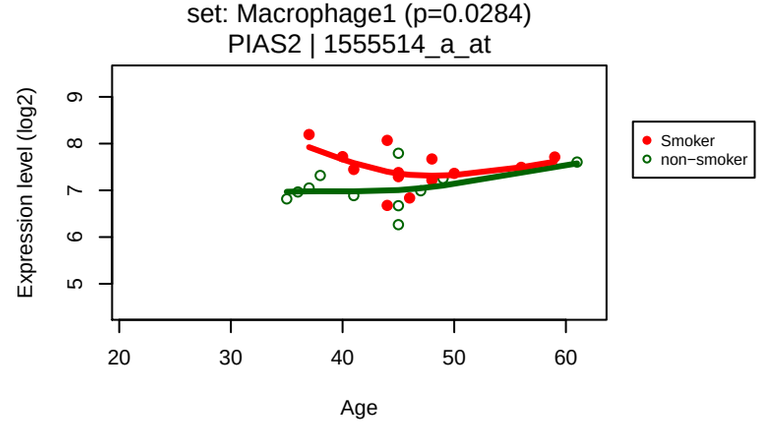

set: SmallAirway1 (p=0.1147)
PIAS2 | 1555514_a_at
Expression level (log2)
Age
Smoker
non−smoker

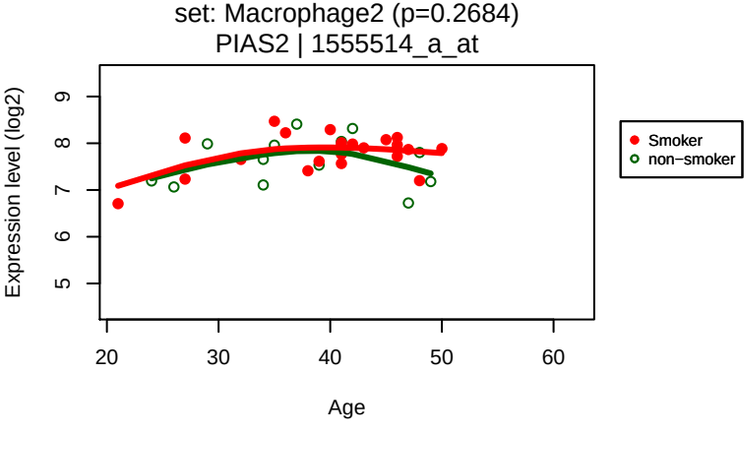

set: SmallAirway2 (p=0.4518)
PIAS2 | 1555514_a_at
Expression level (log2)
Age
Smoker
non−smoker

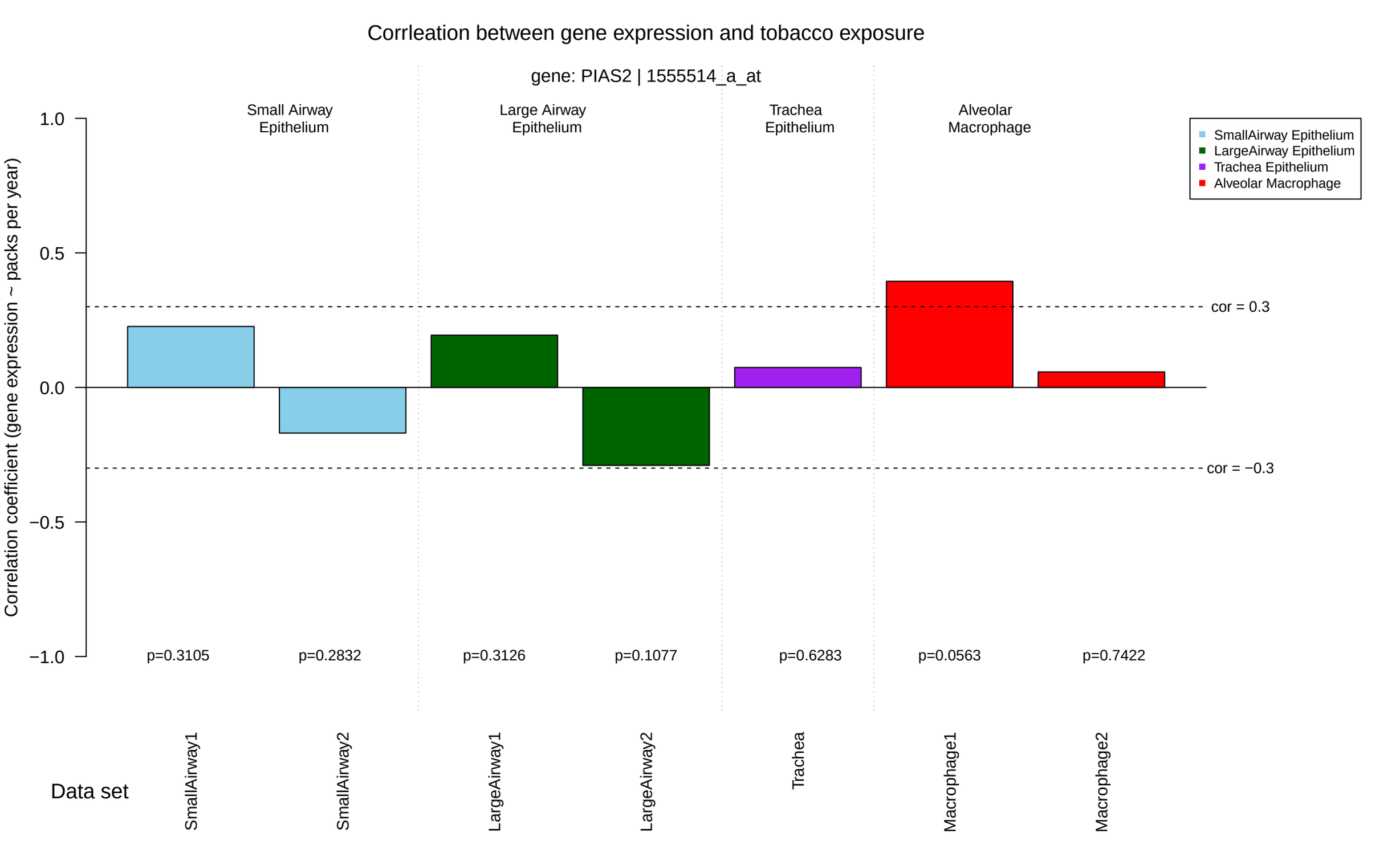

set: LargeAirway1 (p=0.8711)
PIAS2 | 1555514_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5351)
PIAS2 | 1555514_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9253)
PIAS2 | 1555514_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0284)
PIAS2 | 1555514_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2684)
PIAS2 | 1555514_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PIAS2 | 1555514_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3105
p=0.2832
p=0.3126
p=0.1077
p=0.6283
p=0.0563
p=0.7422
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2